

Perceptron Convergence Proof

Suppose we have a binary classification dataset $\{\mathbf{x}^i, y^i\}$, $y_i \in [0, 1]$. We are going to assume that the dataset is finite (i.e. $\max_i \|\mathbf{x}_i\|_2 = R$) and that is linearly separable with margin γ .

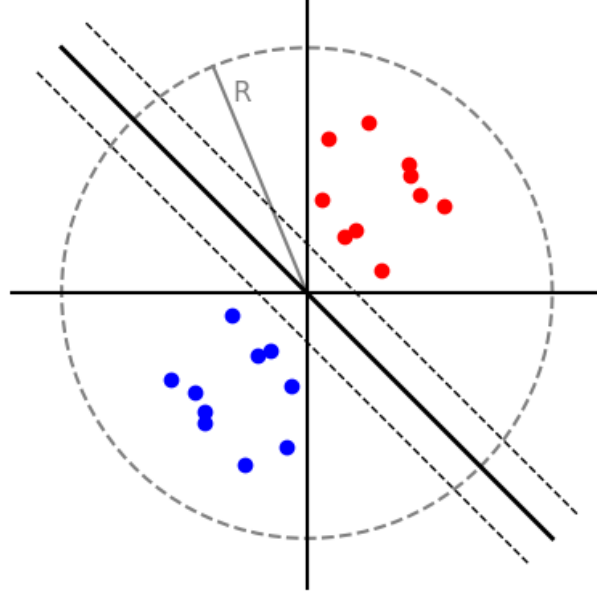


Figure 1: Linearly separable dataset

Linear separability here implies that there exists some “ideal” unit weight vector $\mathbf{w}^*$ such that when $y = 1$, $\mathbf{w}^* \cdot \mathbf{x} \geq \gamma$ and when $y = 0$, $\mathbf{w}^* \cdot \mathbf{x} \leq -\gamma$.

To prove convergence, we will consider a weight update after we have made a mistake. Suppose that we start with $\mathbf{w}_0 = 0$. At step n , $y = 1$ and we predict $\hat{y} = 0$. We perform a weight update

$$\begin{aligned}\mathbf{w}^n &= \mathbf{w}^{n-1} + \alpha(y - \hat{y})\mathbf{x} \\ &= \mathbf{w}^{n-1} + \alpha \mathbf{x}\end{aligned}$$

First, we’ll compute the “progress” towards our ideal weight vector $\mathbf{w}^*$ using a dot product

$$\begin{aligned}\mathbf{w}^* \cdot \mathbf{w}^n &= \mathbf{w}^* \cdot \mathbf{w}^{n-1} + \alpha \mathbf{w}^* \cdot \mathbf{x} \\ &\geq \mathbf{w}^* \cdot \mathbf{w}^{n-1} + \alpha \gamma\end{aligned}$$

Then via induction we can write

$$\mathbf{w}^* \cdot \mathbf{w}^n \geq n \alpha \gamma \tag{1}$$

This lower bound occurs because for each mistake we make, we get at least a little bit closer to our target $\mathbf{w}^*$. We can use this to put a lower bound on the magnitude of $\mathbf{w}^n$

$$\|\mathbf{w}^n\| \geq \mathbf{w}^n \cdot \mathbf{w}^* \geq n \alpha \gamma$$

Next, we will put an upper bound on the same vector using the finiteness of the dataset.

$$\begin{aligned}\|\mathbf{w}^n\|^2 &= \|\mathbf{w}^{n-1} + \alpha \mathbf{x}\|^2 \\ &= \|\mathbf{w}^{n-1}\|^2 + 2\alpha \mathbf{w}^{n-1} \cdot \mathbf{x} + \alpha^2 \|\mathbf{x}\|^2 \\ &\leq \|\mathbf{w}^{n-1}\|^2 + 2\alpha \mathbf{w}^{n-1} \cdot \mathbf{x} + \alpha^2 R^2 \\ &\leq \|\mathbf{w}^{n-1}\|^2 + \alpha^2 R^2\end{aligned}$$

In the final step, we've used the fact that since we made a "mistake" and erroneously predicted $\hat{y} = 0$, $\mathbf{w}^{n-1} \cdot \mathbf{x} \leq 0$. From this we can define an upper bound via induction

$$\|\mathbf{w}^n\|^2 \leq n \alpha^2 R^2 \tag{2}$$

Putting our bounds (1-2) together, we have

$$n^2 \alpha^2 \gamma^2 \leq \|\mathbf{w}^n\|^2 \leq n \alpha^2 R^2$$

$$\Rightarrow n \leq R^2 / \gamma^2$$

Thus, the total number of mistakes we will make in the learning process is at most R^2 / γ^2 .